

How bias arises in Gnocchi

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1 Expected number of SNVs under neutrality

In a selectively neutral genomic window, the expected number of sites that are polymorphic (a site is polymorphic if an ALT allele is seen at that site in one or more members of a given cohort) is

$$\sum_i p_{c_i}(x_i) \tag{1}$$

where:

- i indexes the single-nucleotide sites in the window
- x_i is a vector of regional features (GC content, etc) associated with site i ,
- c_i is the sequence and methylation context of the site i ,
- $p_c(x)$ is the probability that a site is polymorphic, conditional on context c and regional feature vector x , and, of course, the number of individuals in the cohort.

2 Assuming uniformity of the features within a genomic window

Assume that all the features vectors $\{x_i\}$, ranging over all sites i in a window w , are approximately the same:

$$x_i = x_w.$$

Under this assumption, equation (1) simplifies to

$$\sum_i p_{c_i}(x_w) = \sum_c n_c^{(w)} p_c(x_w), \tag{2}$$

where the right side groups probabilities that appear on the left side by context, and $n_c^{(w)}$ is the number of sites in window w that have sequence and methylation context c .

3 The naive way to fit a model of the polymorphism probability

To predict the SNV count of a window with feature vector x , we need to model $p_c(x)$, the probability that a site with context c and feature vector x is polymorphic (as opposed to monomorphic) across the cohort of interest. The natural approach is to construct a parametric model of $p_c(x)$ as follows:

$$p_c(x) = \sigma(\beta_c \cdot x), \quad (3)$$

where:

- $\sigma(x)$ is the sigmoid function (bounded between 0 and 1),
- $\cdot$ is the dot product,
- β_c is a parameter vector associated with the sequence and methylation context c ,

and then train one model for each context c on an appropriate genome-wide training set of sites (each with context c , but with varying feature vectors x) that are either monomorphic or polymorphic in the cohort.

4 The training set is contaminated with selection that may correlate with x

The problem with this approach lies in choosing the training set of sites. These sites should be selectively neutral, but, in practice we don't know how to identify such sites in the noncoding genome, where functional annotations are infrequent and imprecise. Moreover, noncoding sites under selection may correlate with x (e.g., promoters, which are under selection, tend to have high GC content), causing $p_c(x)$ to be inaccurate in the tails of the x -distribution.

5 A 2-stage approach to eliminate the effect of feature-dependent contamination of the training set

De novo mutations (DNMs), by definition, have not been exposed to natural selection, and therefore inform us about the inherent mutability of a genomic site. Can we use DNMs as a training set? Not immediately: we want a neutral estimate of the polymorphism probability, which scales with cohort size, not DNM rate, which does not.

There is a way, however, to draw that DNM rate into the polymorphism probability such that it eliminates the effect of feature-dependent contamination with selection that would otherwise occur in a training set of polymorphisms

constructed from the entire noncoding genome. To leverage a set of neutrally pure DNMs, Chen et al broke the problem into two steps:

$$p_c(x) = r_c(x) p_c(\bar{x}), \quad (4)$$

where $\bar{x}$ is the average of x over all windows of interest, and

$$r_c(x) = \frac{p_c(x)}{p_c(\bar{x})}, \quad (5)$$

is a feature-dependent multiplicative adjustment to the feature-independent $p_c(\bar{x})$.

In the first step, they approximated $p_c(\bar{x})$ by the maximum-likelihood estimate:

$$p_c(\bar{x}) \approx \frac{N_c^{(p)}}{N_c^{(m)} + N_c^{(p)}} \quad (6)$$

where

- $N_c^{(m)}$ is the number of sites in the noncoding genome with context c that are monomorphic, and
- $N_c^{(p)}$ is the number of sites in the noncoding genome with context c that are polymorphic

Given the large number of training examples, and the fact that most of them have feature vectors close to $\bar{x}$, this estimate is likely to be quite accurate (though subject to low levels of contamination by selection).

In the second step, they assumed that the polymorphism probability in the cohort, $p_c(x)$, is approximately proportional to the DNM probability, $\hat{p}_c(x)$, in an arbitrary training set of DNMs and background sites (not DNMs). Under this assumption:

$$r_c(x) \approx \frac{\hat{p}_c(x)}{\hat{p}_c(\bar{x})} \quad (7)$$

where

$$\hat{p}_c(x) = \sigma(\hat{\beta}_c \cdot x), \quad (8)$$

with parameters $\hat{\beta}_c$ estimated from the training set of DNMs and background sites (not DNMs). Notice that, even though $\hat{p}_c(x)$ is sensitive to the class balance in the training set (the ratio of DNMs examples to non-DNM examples), the estimate of $r_c(x)$, which is a ratio of such probabilities, is not. Though this training set can be trusted to be neutral, even in the tails of the x -distribution, it is much smaller than that used to compute $p_c(\bar{x})$, so the estimate of $r_c(x)$ is likely to be noisier than that of $p_c(\bar{x})$ solely because of sampling noise.

6 Hypothesis: the feature-dependent adjustment is biased

As we show in our manuscript, the process of training any function of x works, in general, to minimize global bias at the expense of local bias in the tails of the x -distribution. Applied to the adjustment factor $r_c(x)$, whose training data is sparse, we expect $r_c(x)$ to be accurate near $\bar{x}$, but inaccurate (and possibly biased) in the tails (x far from $\bar{x}$). This bias would propagate, via $p_c(x)$, to the expected per-window SNV counts as a function of x , potentially explaining the bias we observe in Gnocchi at extreme GC content, for example.

7 Predictions of the hypothesis

It is instructive to consider what would happen, under the above hypothesis, as the size of the DNM training set is increased.

When the DNM training set is so small that the tails are not sampled at all, then $r_c(x)$ is expected to tend towards its value at $\bar{x}$, which is 1 by construction, and Gnocchi reduces to a context-only model, inheriting whatever bias it has as a function of x . **We performed this experiment and found not only that bias in Gnocchi was reduced, but also that it approached the levels of bias we saw in other context-only constraint metrics.**

As the DNM training set size increases, and more and more extreme values of x are sampled, $r_c(x)$ will start to capture the true function near $\bar{x}$ making Gnocchi better there, but this may come at the expense of deviating from the true function away from $\bar{x}$, in a way that we cannot predict without knowing the truth. **This is what we believe is causing the large bias we observe in Gnocchi.**

Eventually, if enough DNM training data existed to densely sample extreme x , then we would expect the bias at extreme x to disappear. **We approximated this regime by increasing the number of background sites (only) in the DNM training set, retrained $r_c(x)$, and observed that bias below the level we reported in our paper.**

Together these experiments lend support to our hypothesis that the sparseness of the DNM training set is responsible for the large bias observed in Gnocchi as a function of GC content.

8 Derivation of the quantities plotted in Figure 3

Figure 3 plots three quantities, all binned on the GC content of a 1 kb window. This section derives each of them from the model above, and states exactly which approximations enter and which do not.

8.1 Notation

Write the context $c = (t, m)$, where t is the trinucleotide and m the methylation level. This distinction matters here, and is not cosmetic: step 1 conditions on both t and m , whereas the fitted adjustment of step 2 conditions on t only. Chen et al. fit one model per trinucleotide, pooling all methylation levels.

Let W be the analyzed set of N windows (noncoding, passing quality control, autosomes and pseudoautosomal regions), let g index a GC bin, and let $W_g \subset W$ be the windows falling in bin g . For a window w , let $O(w)$ be the observed number of polymorphic sites in the cohort, and $n_c^{(w)}$ the number of callable sites of context c in w , as before.

Two further points of fact about the fitted model, both confirmed against Chen et al.’s code rather than their text. First, the operative adjustment is

$$r_t(x) = \frac{\sigma(\beta_{t,0} + \beta_t \cdot z(x))}{\sigma(\beta_{t,0})}, \quad (9)$$

a ratio of two *predicted probabilities* from the same fitted model, with $z(x)$ the standardized, PCA-rotated feature vector and $z(\bar{x}) = 0$ by construction. (Their Methods state a ratio of linear predictors instead; the code computes Eq. (9).) Second, and consequentially, r_t is a *ratio*: any error in $\hat{p}_t$ that is a pure level shift, common to numerator and denominator, cancels exactly and never reaches the constraint score. Only feature-driven *variation* in $\hat{p}_t$ survives. Everything below is organized around that fact.

8.2 The two expected counts

The context-only (“step 1”) expected count of a window is Eq. (1) with the adjustment set to unity:

$$E_1(w) = \sum_c n_c^{(w)} p_c(\bar{x}), \quad E_1^t(w) = \sum_{c: t(c)=t} n_c^{(w)} p_c(\bar{x}), \quad (10)$$

so that $E_1 = \sum_t E_1^t$. The adjusted (“step 2”, published Gnocchi) expected count applies r per trinucleotide:

$$E_2(w) = \sum_t E_1^t(w) r_t(x_w). \quad (11)$$

Both are available per window and per (w, t) from the pipeline’s own exports, so neither is reconstructed here.

8.3 Panel A: the mean standardized rank

For a model $M \in \{1, 2\}$ Chen et al. summarize a window by the signed chi statistic $z_M(w) = -\text{sign}(O - E_M) \sqrt{(O - E_M)^2 / E_M}$, which is just

$$z_M(w) = \frac{E_M(w) - O(w)}{\sqrt{E_M(w)}}, \quad (12)$$

positive when a window carries less variation than expected. Windows are retained only if $z_M \in [-10, 10]$ for every M plotted, so all curves describe one identical window set. Each z_M is then converted to a standardized rank within that set,

$$u_M(w) = \frac{\text{rank}_M(w) - \frac{1}{2}}{N} \in (0, 1), \quad (13)$$

which is uniform on $(0, 1)$ by construction, with mean exactly $1/2$. Panel A plots the conditional mean

$$\bar{u}_M(g) = \frac{1}{|W_g|} \sum_{w \in W_g} u_M(w), \quad (14)$$

with error bars $\text{sd}(u_M)/\sqrt{|W_g|}$. Because the marginal distribution of u_M is uniform regardless of the model, $\bar{u}_M(g) = 1/2$ for every g is exactly the statement that the model has no GC-dependent bias, and departures from $1/2$ measure that bias on a scale that is comparable across models with entirely different units. This is also why the third curve, depletion rank, can be overlaid despite being defined on a different (Halldorsson) window set: it is ranked within itself, and the sign convention is opposite, so its complement $1 - \text{DR}$ is plotted.

8.4 Why panel A is so sensitive to the adjustment

It is worth recording how strongly Eq. (14) responds to r , because this is what licenses reading panel B as the cause of panel A. Suppose a window's adjustment is a uniform inflation, $E_2 = fE_1$, and write the observed count as $O = E_1(1 + \epsilon)$. Then from Eq. (12),

$$z_2 - z_1 = \sqrt{E_1} \left[\frac{f - 1 - \epsilon}{\sqrt{f}} + \epsilon \right] \approx \sqrt{E_1} (f - 1) \quad (f \rightarrow 1, \epsilon \rightarrow 0). \quad (15)$$

The $\sqrt{E_1}$ prefactor is the point. A typical analyzed window has $E_1 \approx 174$, so $\sqrt{E_1} \approx 13$, and a mere 10% inflation displaces z by more than one unit — comparable to the spread of z itself. Measured directly by re-ranking the real genome-wide z distribution under a uniform inflation f , the mean rank moves from 0.500 at $f = 1$ to 0.687 at $f = 1.10$ and 0.302 at $f = 0.90$. Small multiplicative errors in r are therefore not a second-order concern; they are the whole effect.

8.5 Panel B, first form: the adjustment Gnocchi applies

Dividing Eq. (11) by Eq. (10) gives the single multiplicative factor each window actually receives,

$$\bar{r}(w) = \frac{E_2(w)}{E_1(w)} = \sum_t \frac{E_1^t(w)}{E_1(w)} r_t(x_w), \quad (16)$$

an E_1 -weighted mean of the per-trinucleotide adjustments. Aggregating to a GC bin as a ratio of sums rather than a mean of ratios,

$$R(g) = \frac{\sum_{w \in W_g} E_2(w)}{\sum_{w \in W_g} E_1(w)}, \quad (17)$$

which is the adjustment the bin receives in aggregate, and which keeps the following decomposition exact.

Let $K = \{\text{ACG, CCG, GCG, TCG}\}$ be the CpG trinucleotides. Define the CpG share of the step-1 expectation and the two group-wise adjustments,

$$\Pi(g) = \frac{\sum_{w \in W_g} \sum_{t \in K} E_1^t(w)}{\sum_{w \in W_g} E_1(w)}, \quad R_{\text{CpG}}(g) = \frac{\sum_{w \in W_g} \sum_{t \in K} E_1^t(w) r_t(x_w)}{\sum_{w \in W_g} \sum_{t \in K} E_1^t(w)}, \quad (18)$$

and R_{non} likewise over $t \notin K$. Splitting the numerator and denominator of Eq. (17) at K gives, with no approximation whatsoever,

$$R(g) = \Pi(g) R_{\text{CpG}}(g) + (1 - \Pi(g)) R_{\text{non}}(g). \quad (19)$$

The first form of panel B plots the three curves of Eq. (19) against g , together with the counterfactual obtained by holding the non-CpG adjustment at unity,

$$R^{\text{cf}}(g) = \Pi(g) R_{\text{CpG}}(g) + (1 - \Pi(g)). \quad (20)$$

Empirically $R_{\text{CpG}}(g)$ is flat at 0.99–1.00 across the entire GC range while $R_{\text{non}}(g)$ climbs from 0.96 to 1.79, and $R^{\text{cf}}(g)$ is flat to within 0.6% even though Π reaches 0.43 at the highest GC. Since Eq. (19) is an identity, this is not a fit but a bookkeeping statement: the GC dependence of what Gnocchi applies is carried entirely by the non-CpG contexts.

That CpG models are inert is a direct consequence of the ratio structure noted after Eq. (9). Chen et al. deliberately strip GC content, CpG island status, and three correlated features from the CpG-context models; what remains barely varies with GC, so the numerator and denominator of Eq. (9) move together and $r_t \approx 1$. Their large level miscalibration — these models cannot represent methylation, and hypomethylated CpG islands in GC-rich sequence mutate at roughly a third the rate of methylated CpGs — cancels, and never reaches the score. Methylation is thus *ruled out* as a cause of the bias, not implicated in it.

8.6 Panel B, second form: the adjustment the DNMs support

Establishing that the non-CpG adjustment drives the bias does not establish that it is *wrong*. For that we need an empirical counterpart of r_t computed from data the model was meant to describe. Since r_t is by definition the factor by which the true mutation rate departs from its context-average value, the empirical target is a rate, measurable per trinucleotide and GC bin:

$$\tilde{p}_t(g) = \frac{D_t(g)}{A_t(g)}, \quad (21)$$

where $D_t(g)$ is the number of observed de novo mutations of context t falling in W_g , and $A_t(g)$ is the number of *opportunities* — possible SNVs of context t — in the same windows. Numerator and denominator are restricted to the same window set, and DNMs are assigned to windows by position, so the denominator counts exactly the territory the numerator was counted over.

Normalization is forced, not chosen. Chen et al.’s own assumption (Sec. 6) is that the polymorphism and DNM probabilities are proportional, $\hat{p}_t(x) = \lambda_t p_t(x)$, with λ_t unknown and *context-specific*: it absorbs both the cohort-size scaling and the fact that $p_t(\bar{x})$ saturates by different amounts in different contexts. The proportionality constant cancels in a within-context ratio,

$$\frac{\hat{p}_t(x)}{\hat{p}_t(\bar{x})} = \frac{\lambda_t p_t(x)}{\lambda_t p_t(\bar{x})} = r_t(x), \quad (22)$$

and only there. So the empirical adjustment must be formed as the same ratio, each context normalized by its own E_1 -weighted average over bins,

$$\hat{r}_t^{\text{obs}}(g) = \frac{\tilde{p}_t(g)}{\langle \tilde{p}_t \rangle}, \quad \langle \tilde{p}_t \rangle = \frac{\sum_{g'} E_1^t(g') \tilde{p}_t(g')}{\sum_{g'} E_1^t(g')}, \quad (23)$$

where $E_1^t(g) = \sum_{w \in W_g} E_1^t(w)$. No free constant is introduced and none is fitted; λ_t never has to be estimated.

The model side must then be normalized *identically*, $\hat{r}_t^{\text{mod}}(g) = r_t(g)/\langle r_t \rangle$ with $r_t(g) = E_2^t(g)/E_1^t(g)$. This is not symmetry for its own sake. As GC increases the trinucleotide composition of windows shifts; if only the empirical side were normalized, the residual between-context level differences on the model side would be reweighted bin by bin and would appear in the aggregate as GC dependence that no model actually asserts.

The two curves of the second form of panel B are then the same E_1 -weighted aggregate over non-CpG contexts,

$$R_{\text{non}}^\bullet(g) = \frac{\sum_{t \notin K} E_1^t(g) \hat{r}_t^\bullet(g)}{\sum_{t \notin K} E_1^t(g)}, \quad \bullet \in \{\text{mod}, \text{obs}\}, \quad (24)$$

and the lower sub-panel plots their ratio, the over-adjustment

$$\Phi(g) = \frac{R_{\text{non}}^{\text{mod}}(g)}{R_{\text{non}}^{\text{obs}}(g)}. \quad (25)$$

A bin’s contribution is used only if it carries enough observed DNMs to estimate Eq. (21), and a bin is reported only if the usable contexts still account for at least 90% of its non-CpG E_1 weight, so that no bin is built from an unrepresentative minority of contexts. The error bar is Poisson on $D_t(g)$, propagated through Eqs. (23)–(25); the model curve carries none, being a deterministic function of already-fixed feature values.

Measured this way, $R_{\text{non}}^{\text{mod}}$ climbs monotonically from 0.95 to 1.55 while $R_{\text{non}}^{\text{obs}}$ stays near unity until GC ≈ 0.55 , so Φ crosses 1 near GC = 0.40 and reaches 1.22–1.26, many standard errors from unity. By Eq. (15) an over-adjustment of that size is more than enough to produce the rank displacement in panel A.

Why opportunities, and not expected counts, in Eq. (21). A rate needs a count of opportunities in its denominator. It happens that the step-1 expected count would serve equally well here, because a non-CpG trinucleotide has a single methylation level and therefore a single set of fitted per-site rates:

$$E_1^t(g) = \sum_{\text{alt}} A_t^{\text{alt}}(g) \pi_t^{\text{alt}} = A_t(g) \mu_t, \quad \mu_t = \frac{1}{3} \sum_{\text{alt}} \pi_t^{\text{alt}}, \quad (26)$$

with $\pi_t^{\text{alt}} = p_c(\bar{x})$ the fitted per-site rate, and where the second equality uses $A_t^{\text{alt}}(g) = A_t(g)/3$: every callable position of context t contributes one opportunity per alternative allele, so each of the three alleles carries the same site count. The per-context constant μ_t is absorbed by the normalization Eq. (23), so the two denominators give the same curve — verified to agree within 0.1% at every bin. Replacing $A_t(g)$ instead by a real sample of matched background sites from the DNM training set, which removes the fitted model from the denominator altogether, changes the curve by 1–2%. The result does not rest on the choice.

The one assumption that does bite: callability. $D_t(g)$ counts DNMs anywhere in a window, whereas $A_t(g)$ counts only positions gnomAD can call. That fraction is not flat in GC — short-read coverage degrades in GC-rich sequence, and the callable fraction $\gamma(g)$ falls from 0.91 at GC = 0.30 to 0.75 at GC = 0.61. If trio DNM ascertainment tracks gnomAD’s, the two are matched and no correction applies. If instead the trio call sets are closer to complete, the matched denominator is $A_t(g)/\gamma(g)$, so the corrected empirical rate is $\gamma(g) \tilde{p}_t(g)$: since γ decreases with GC, the corrected empirical curve falls further in the tail and Φ *grows*, from 1.22 to 1.44 at GC = 0.61. The finding survives either way and the correction only strengthens it, so the honest statement is a range, $\Phi \in [1.22, 1.44]$, rather than a point estimate. The residual, untestable possibility runs the other way: if trio DNM calling is itself *less* sensitive in GC-rich sequence than gnomAD is, part of the measured over-adjustment is technical rather than a property of the model.

9 A proposal for Gnocchi 2.0

Though DNMs ameliorate the effect of selection contamination of the training set, the trade-off is that the sparseness of DNMs introduces bias into the constraint metric, precisely where it was hoped DNMs would help. Perhaps the quest to avoid contamination of the training set with sites under selection was misguided, and it is better to tolerate some contamination if it yields a less biased constraint score where it matters most. This suggests an approach to a possibly improved constraint score: capture the x dependence in Eq. (3) by training directly on dense gnomAD data, instead of factorizing it and training one of the factors on sparse DNM data. Since this requires a new training process, we deemed this experiment out-of-scope for this rebuttal.